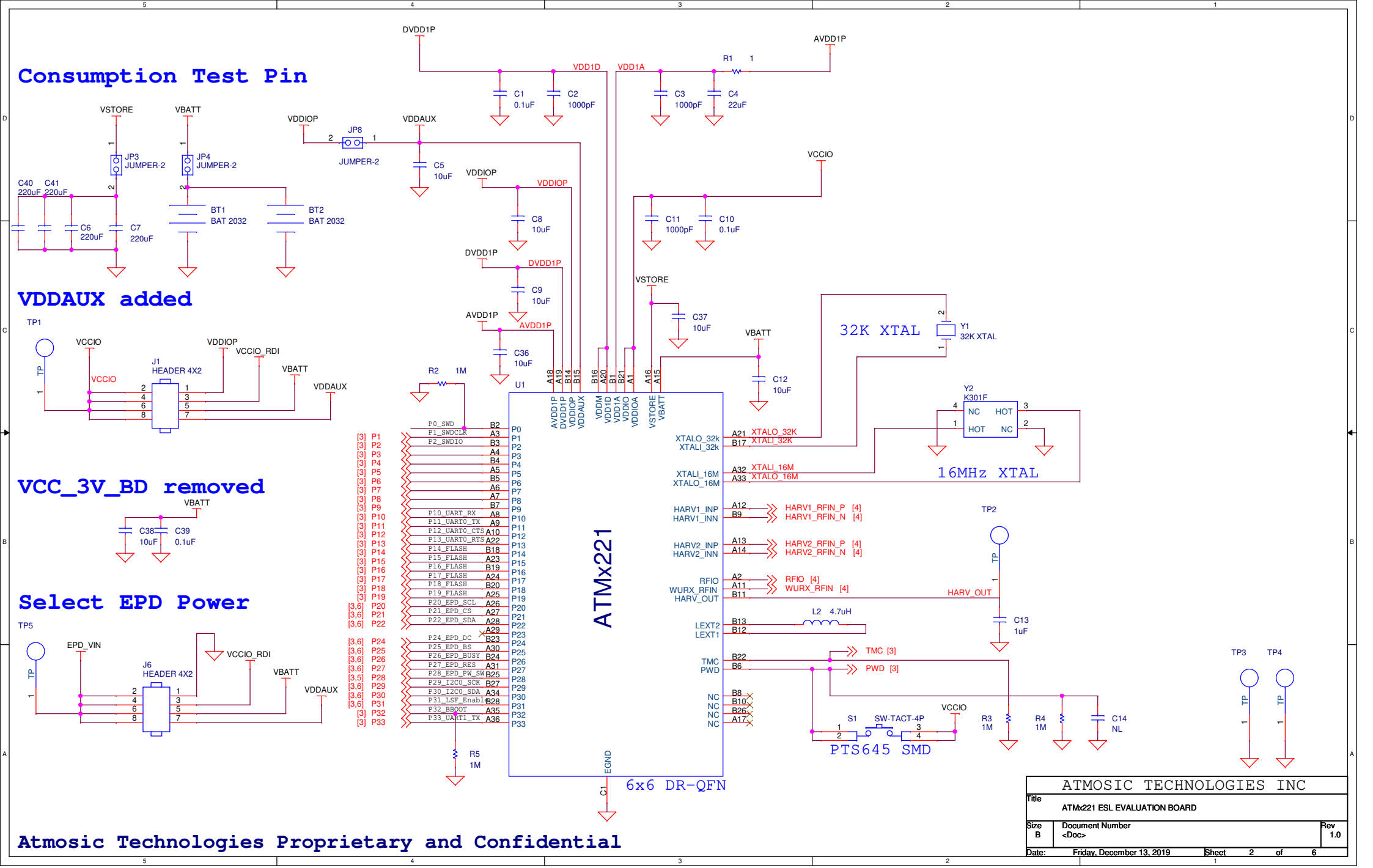
ATMx221 ESL REFERENCE DESIGN BOARD

Release Date	Revision	Design Eng.	Design Note
06/12/2019	Rev. 1.0	Afan	Initial release

Atmosic Technologies, Inc
910 E. Hamilton Ave, Suite 550
Campbell, CA 95008, USA

ATMOSIC TECHNOLOGIES INC			
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Date: Friday, December 13, 2019	Sheet 1 of 6		

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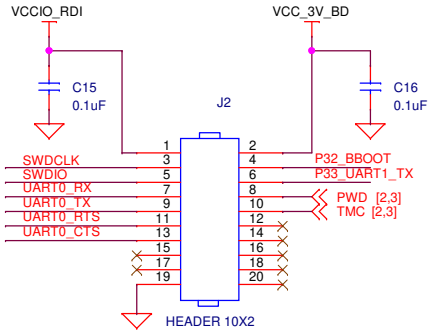

Consumption Test Pin
VSTORE
VBATT
VDDIOP
VDDAUX
JP3 JUMPER-2
JP4 JUMPER-2
JP8 JUMPER-2
C40 220uF
C41 220uF
C6 220uF
C7 220uF
BT1 BAT 2032
BT2 BAT 2032
C5 10uF
DVDD1P
AVDD1P
VDD1D
VDD1A
C1 0.1uF
C2 1000pF
C3 1000pF
C4 22uF
R1 1
VCCIO
C11 1000pF
C10 0.1uF
C8 10uF
C9 10uF
C36 10uF
C37 10uF
C12 10uF
VDDAUX added
TP1
J1 HEADER 4X2
VCCIO_RDI
R2 1M
U1
ATMx221
6x6 DR-QFN
32K XTAL
Y1 32K XTAL
Y2 K301F
16MHz XTAL
XTALO_32k
XTALI_32k
XTALI_16M
XTALO_16M
HARV1_INP
HARV1_INN
HARV2_INP
HARV2_INN
RFIO
WURX_RFIN
HARV_OUT
LEXT2
LEXT1
L2 4.7uH
TMC
PWD
NC
VCC_3V_BD removed
C38 10uF
C39 0.1uF
Select EPD Power
TP5
EPD_VIN
J6 HEADER 4X2
TP2
C13 1uF
S1 SW-TACT-4P
PTS645 SMD
R3 1M
R4 1M
C14 NL
TP3
TP4
R5 1M
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GPIOs SELECTION

[2] P1 — SWDCLK
[2] P2 — SWDIO
[2] P10 — UART0_RX
[2] P11 — UART0_TX
[2] P12 — UART0_CTS
[2] P13 — UART0_RTS
[2] P32 — P32_BBOOT
[2] P33 — P33_UART1_TX
[2] P14 — FLASH_SCLK
[2] P15 — FLASH_CS_L
[2] P16 — FLASH_SIO0
[2] P17 — FLASH_SIO1
[2] P18 — FLASH_SIO2
[2] P19 — FLASH_RESET_L

CONNECTORS

TO INTERFACE BOARD



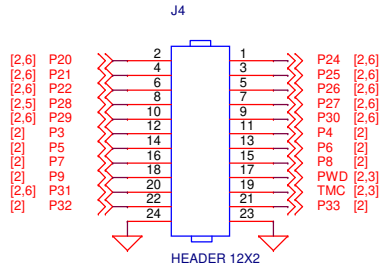
Amphenol Minitek 127 series Receptacle
P/N 20021321-00020C4LF

TO INTERFACE BOARD

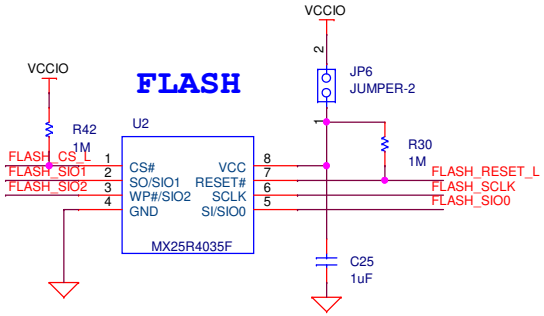
Removed

TO SENSOR HUB etc.

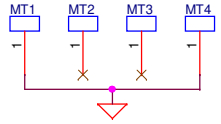
Removed, just signal debug kept



FLASH



MOUNTING HOLES



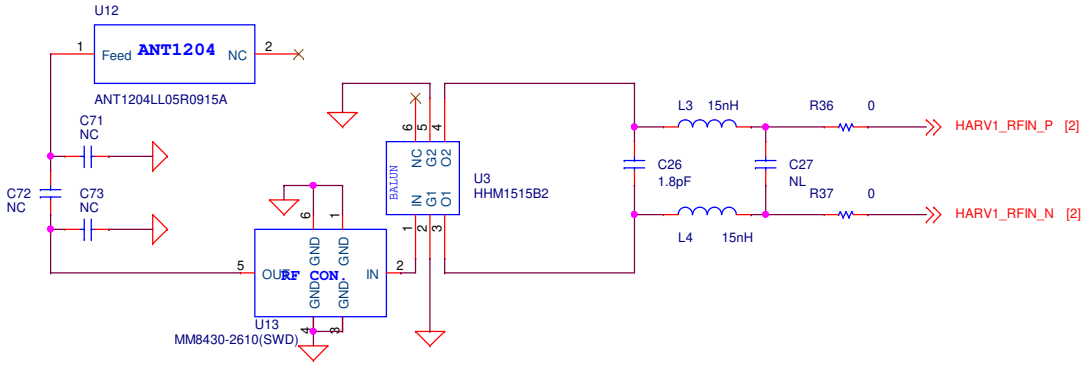
BOARD FIDUCIALS

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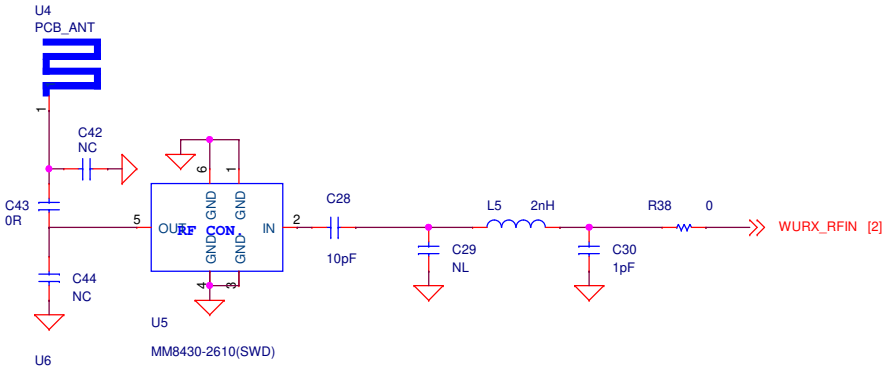
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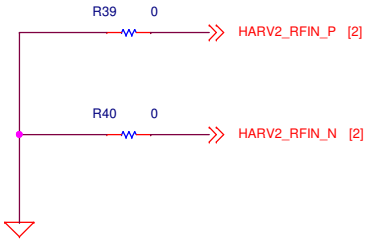
HARV1



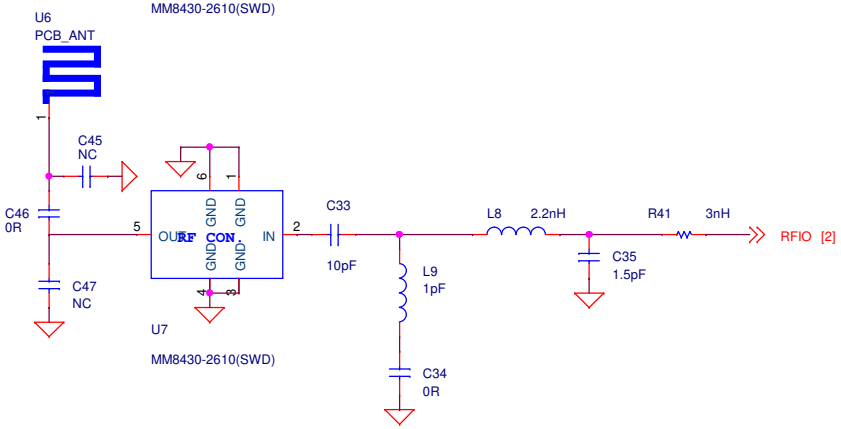
RFIN



HARV2 Removed

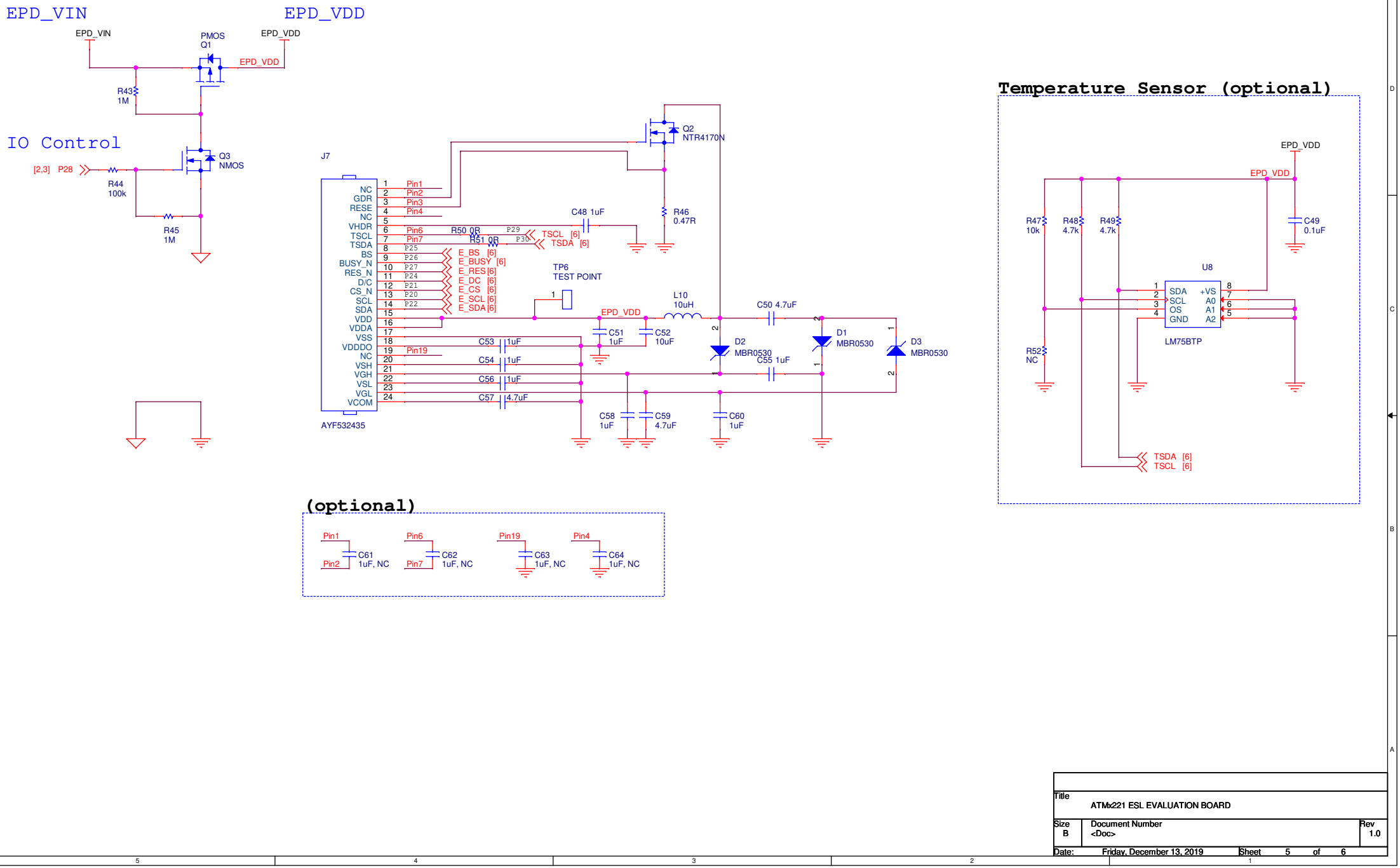


RFIO



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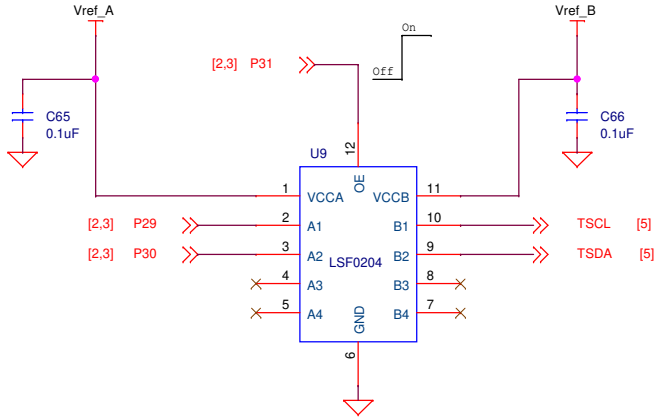
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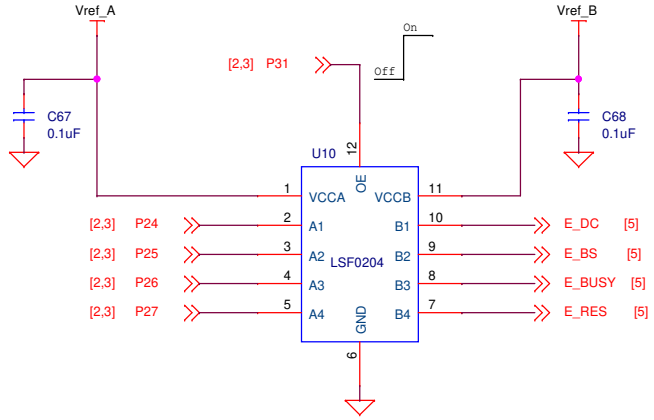


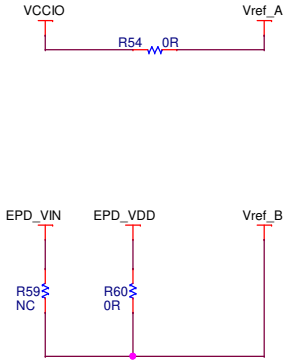
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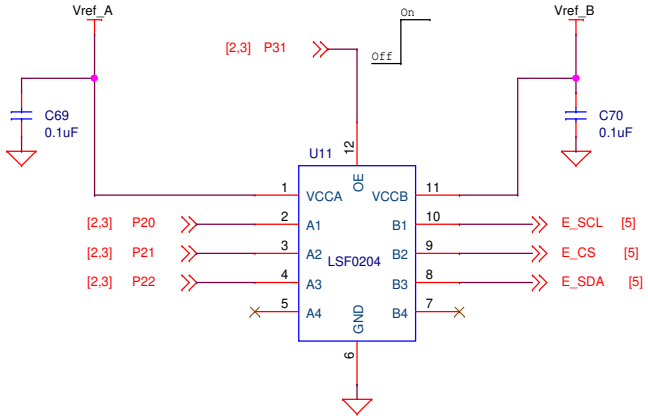
Default : Vref_A is from VCCIO

Default : Vref_B is from EPD_VDD









If VDDIO = EPD_VIN (Optional)

Vref_A is Open

Vref_B is Open

Below all need short wi 0R

[2,3] P20	R55 NC	E_SCL	[5]
[2,3] P21	R56 NC	E_CS	[5]
[2,3] P22	R57 NC	E_SDA	[5]
[2,3] P24	R58 NC	E_DC	[5]
[2,3] P25	R61 NC	E_BS	[5]
[2,3] P26	R62 NC	E_BUSY	[5]
[2,3] P27	R63 NC	E_RES	[5]
[2,3] P29	R64 NC	TSCL	[5]
[2,3] P30	R65 NC	TSDA	[5]

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